



Statistical Methods & Machine Learning 1

Class #3

John Kubale

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Outline

- Review
- Statistical Learning
 - E.g., Studying income
- Linear Models

Review – Population vs. Sample Variance

- Population variance: $\sigma^2 = \frac{\sum_{i=1}^N (y_i - \mu)^2}{N}$
- Sample variance: $s^2 = \frac{\sum_{i=1}^n (y - \hat{y})^2}{n-1}$

Review – Standard Deviation vs. Standard Error

- Population standard deviation: $\sigma = \sqrt{\sigma^2}$
- Population standard deviation: $s = \sqrt{s^2}$
- Standard error: $se = \frac{s}{\sqrt{n}}$

Review – hypothesis testing

- We are often interested in comparing quantities in 2 populations to assess for potential differences (or one population against some known quantity)
- These are generally samples which means there is a certain amount of variability.
 - So, how do we know if there are in fact differences? -> hypothesis testing

Review – Hypothesis testing

- Comparing means of 2 groups (x vs. y)
 - Two-sided test - $H_0: \mu_x = \mu_y$, $H_A: \mu_x \neq \mu_y$
 - One-sided test - $H_0: \mu_x \geq \mu_y$, $H_A: \mu_x < \mu_y$
- How do we know whether we can reject the null hypothesis (H_0)?
 - Look at our p-value -> if lower than our pre-set alpha value (usually 0.05) then we reject the null
 - How/why does this work?

Review – Hypothesis testing

- See “test_stats.pdf” on Canvas.

Review – Alpha and confidence intervals

- We often set $\alpha = 0.05$, but what does this mean?
 - Corresponds to the probability of seeing a value (e.g., mean) as extreme as the value observed if the null hypothesis is true.
- What about 95% confidence intervals (CI)?
 - If we were to calculate a sample mean repeatedly the range represented by our 95% CI would contain the true population mean 95% of the time.

Review – Test assumptions

- There is generally a trade-off between flexibility and interpretability.
- Some tests are more easily interpretable, but to facilitate this require some assumptions.
 - T-tests: Normality, independence*, equal errors*
 - Pearson correlation: Bivariate normality
- If concerned about assumption violation(s) consider using alternatives (e.g., Wilcoxon or Spearman)



Statistical Learning

Y as a function of X

- Y: Outcome, output, dependent, response variable
- X: Feature, input, independent, explanatory, predictor variable(s)
- Learning about Y as a function of X: $Y = f(X) + \epsilon$

What is f ?

What can we do with f ?

Explain Y with X

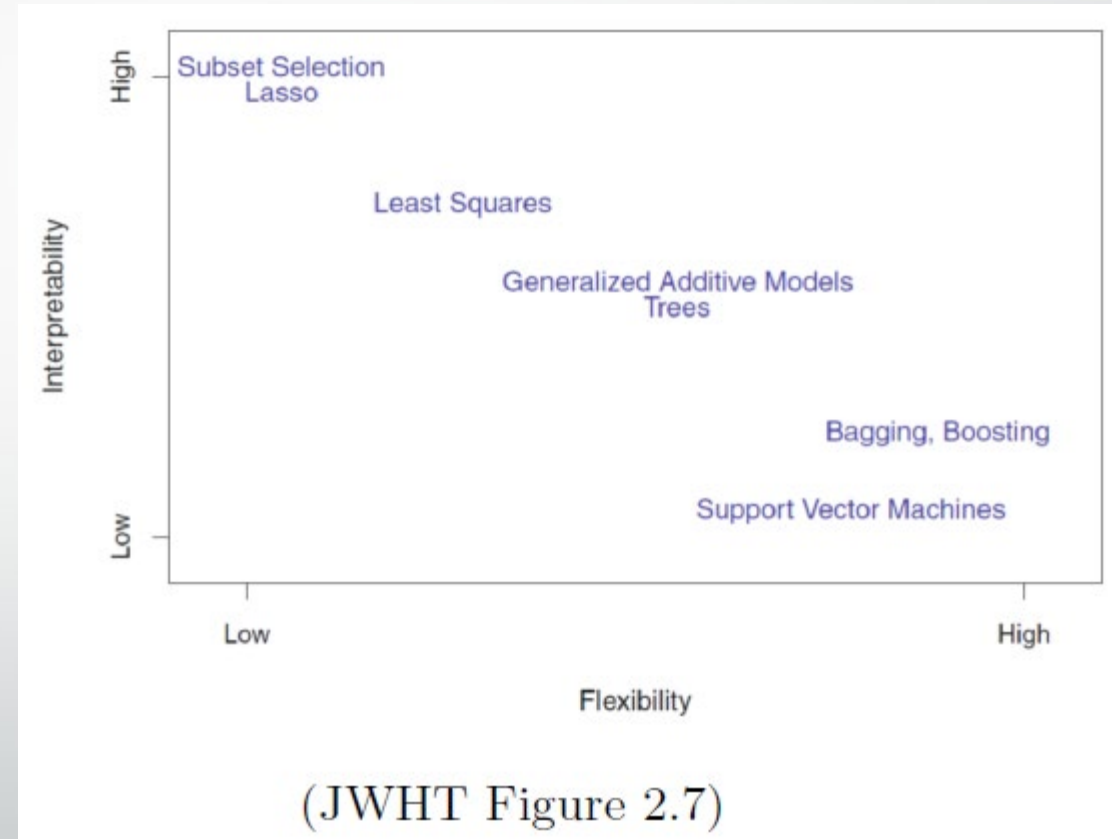
Predict $\hat{Y}$ for specific instances of X

What is ϵ ?

Statistical Learning - 2

How do we find f ?

- Larger $P \rightarrow$ better f
 - *More parameters $\rightarrow$ better fit to data*
- Parametric vs. Non-parametric:
 - Model Interpretability vs. Flexibility

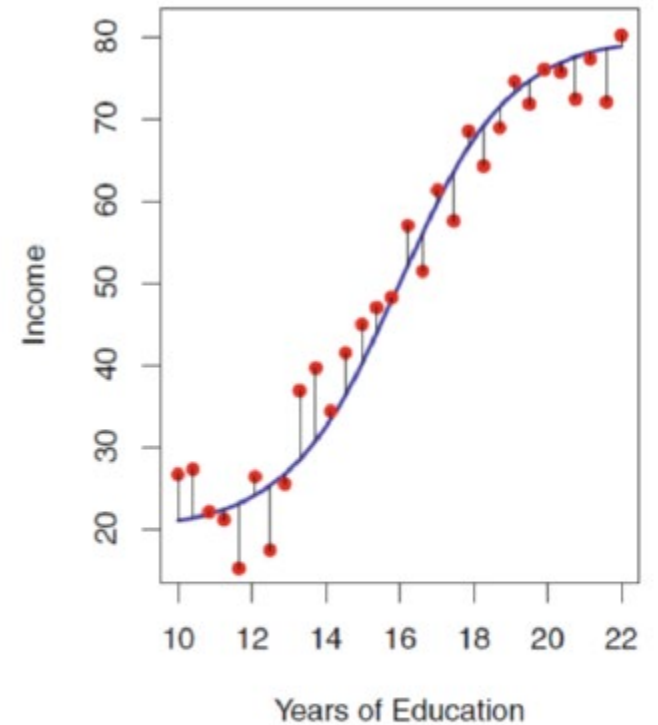
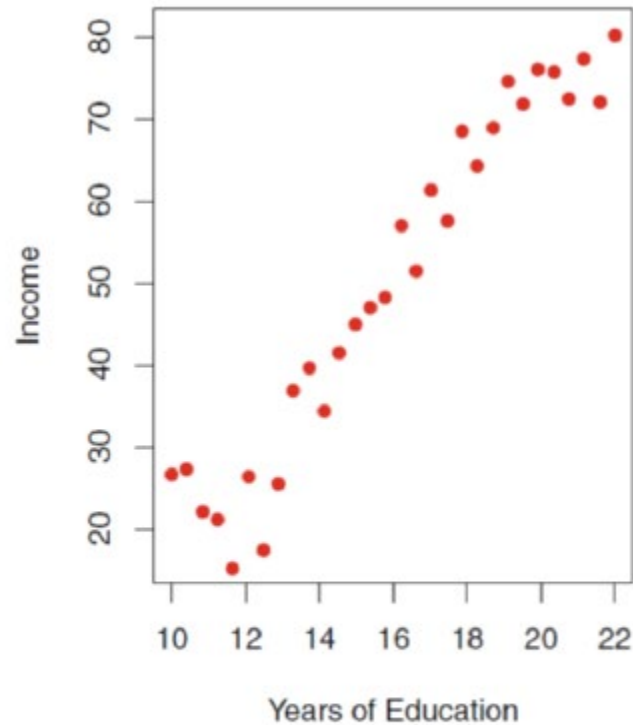




Example: Studying Income

One Predictor

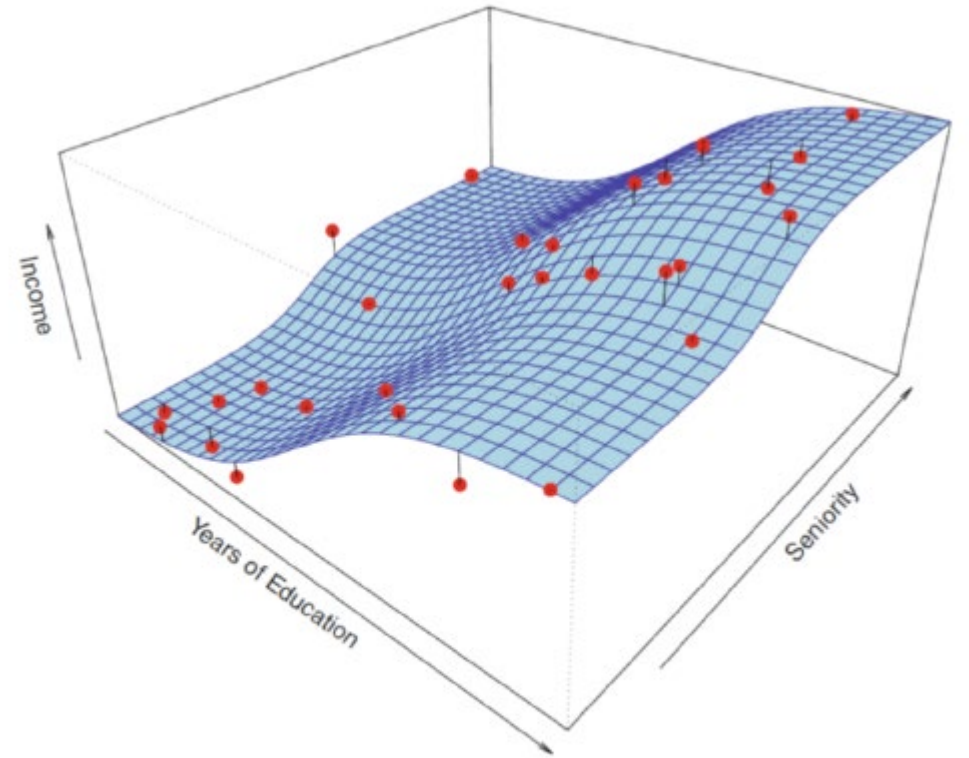
$$\text{Income} = f(\text{Educ}) + \epsilon$$



(JWHT Figure 2.2)

Two Predictors - 1

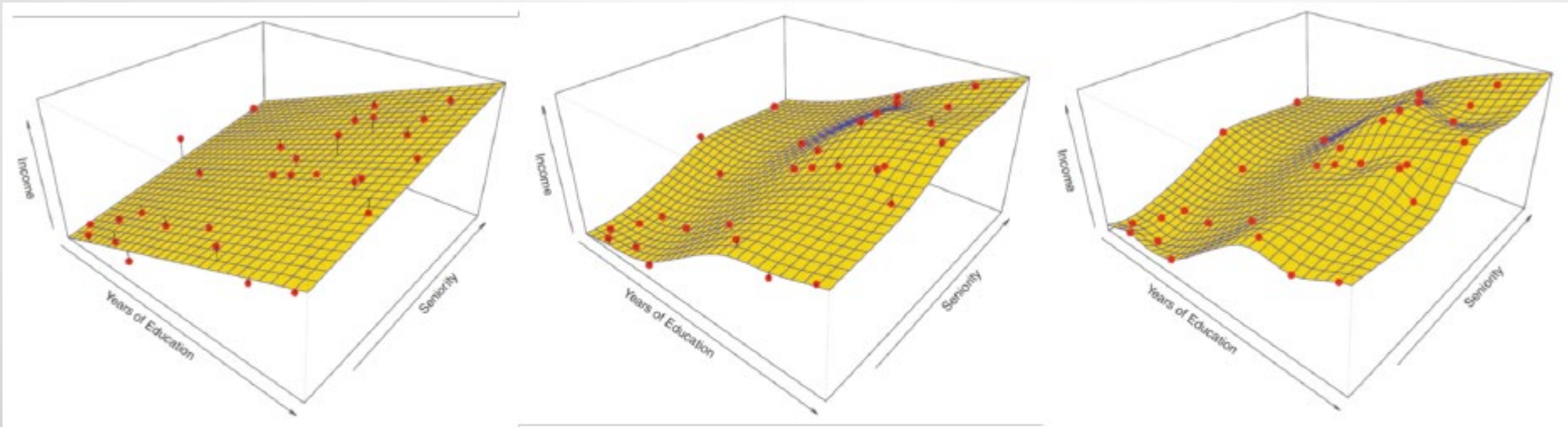
$$\text{Income} = f(\text{Educ}, \text{Seniority}) + \epsilon$$



(JWHT Figure 2.3 - True f)

Two Predictors - 2

$$\text{Income} = f(\text{Educ}, \text{Seniority}) + \epsilon$$



(JWHT Figure 2.4, 2.5, 2.6 - Estimated f)

Which of the 3 figures represents the most interpretable modeling approach?
Which represents the most flexible approach?
Which would you choose? Why?



Linear Models

Linear Regression

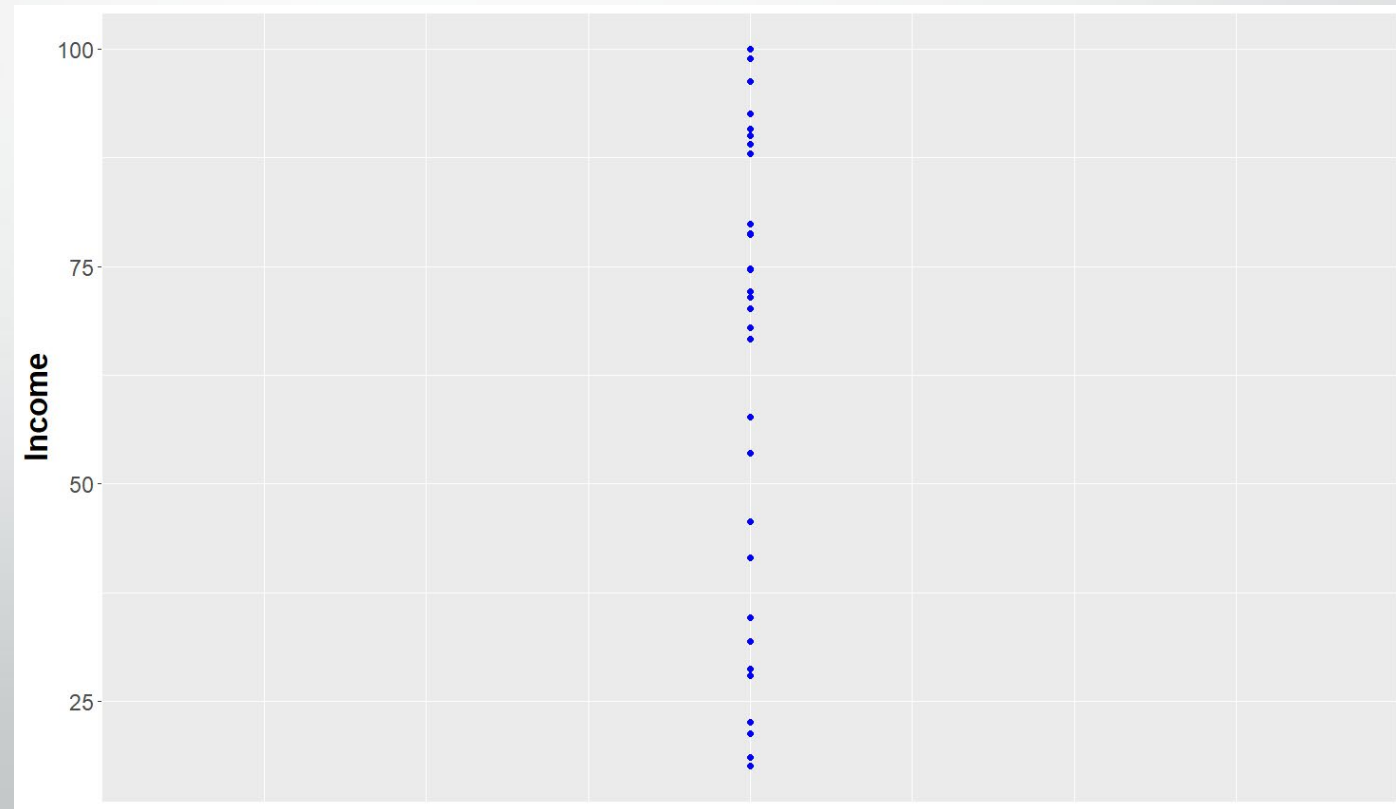
- Linear regression models are one common approach for modeling y as a function of x .

$$Y = \beta_0 + \beta_1 + \epsilon$$

Linear regression with no predictors

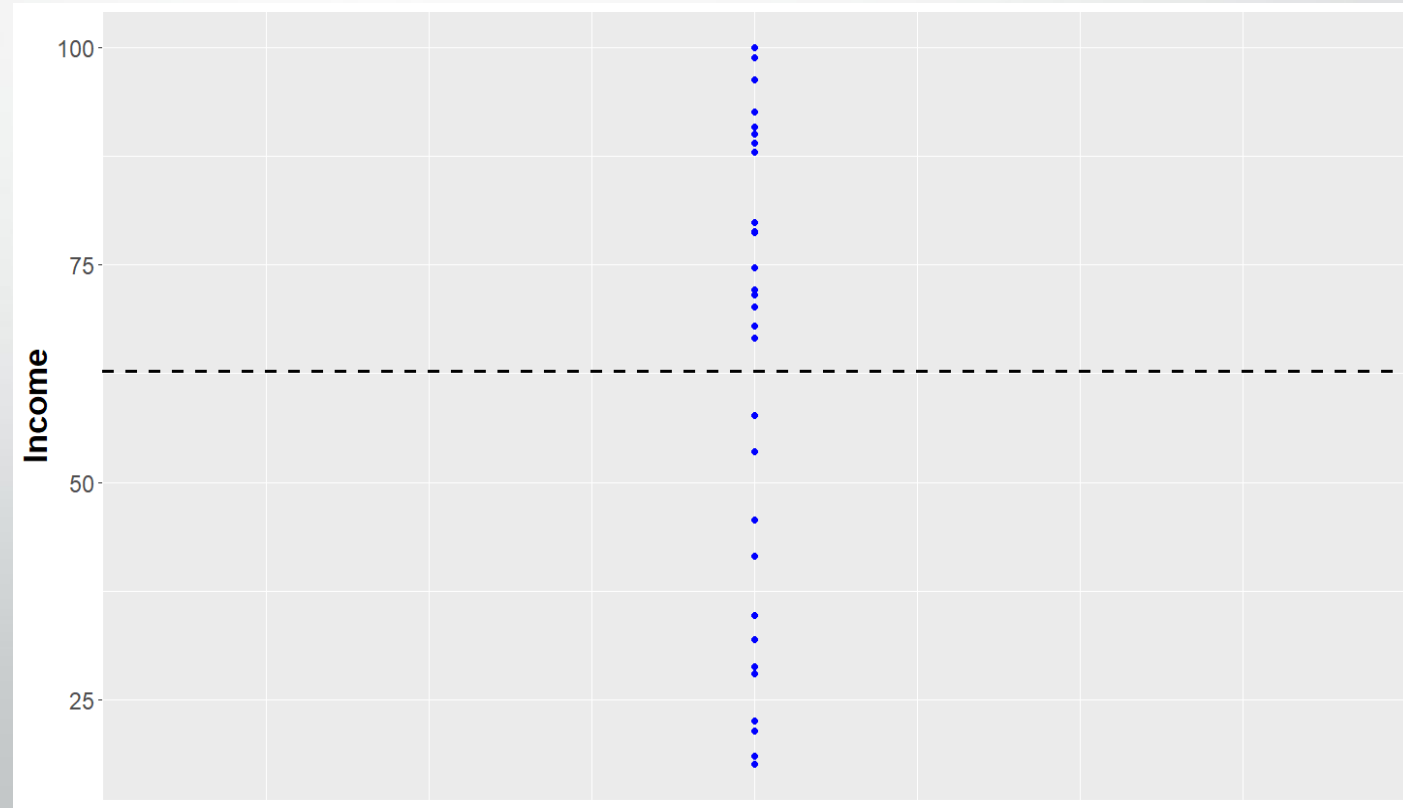
- Using the income2 data we're going to start with the simplest linear regression model – the intercept only model (i.e., no predictors)
 - $E(\text{Income}) = \beta_0 + \epsilon$
- Often written as:
 - $E(\text{Income}) = \bar{Y} + \epsilon$

What is your best guess about where the regression line will be?



Linear regression with no predictors

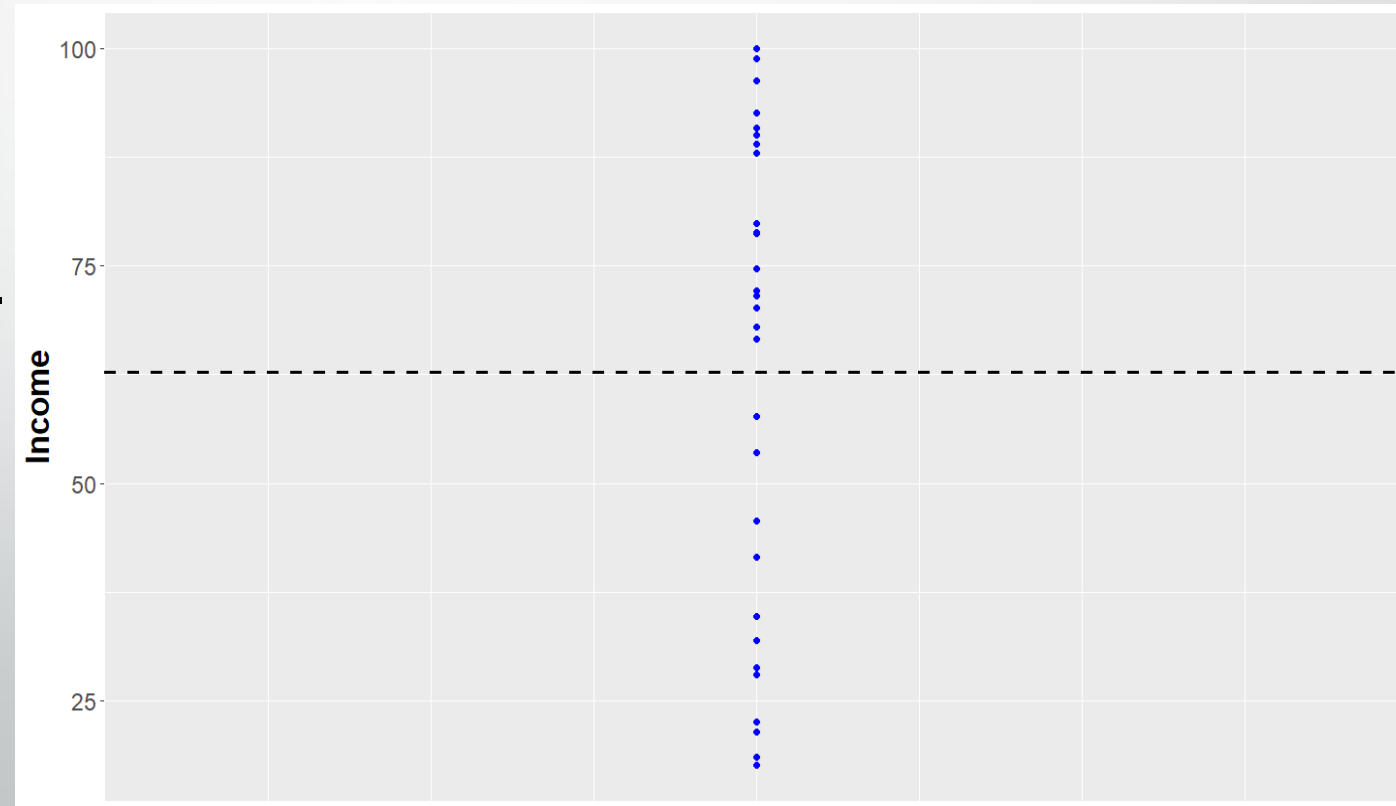
- Let's look at what this looks like graphically.
- What does the dotted line (β_0) represent?



Linear regression with no predictors

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- What does the dotted line (β_0) represent?

It's simply the mean of Income.

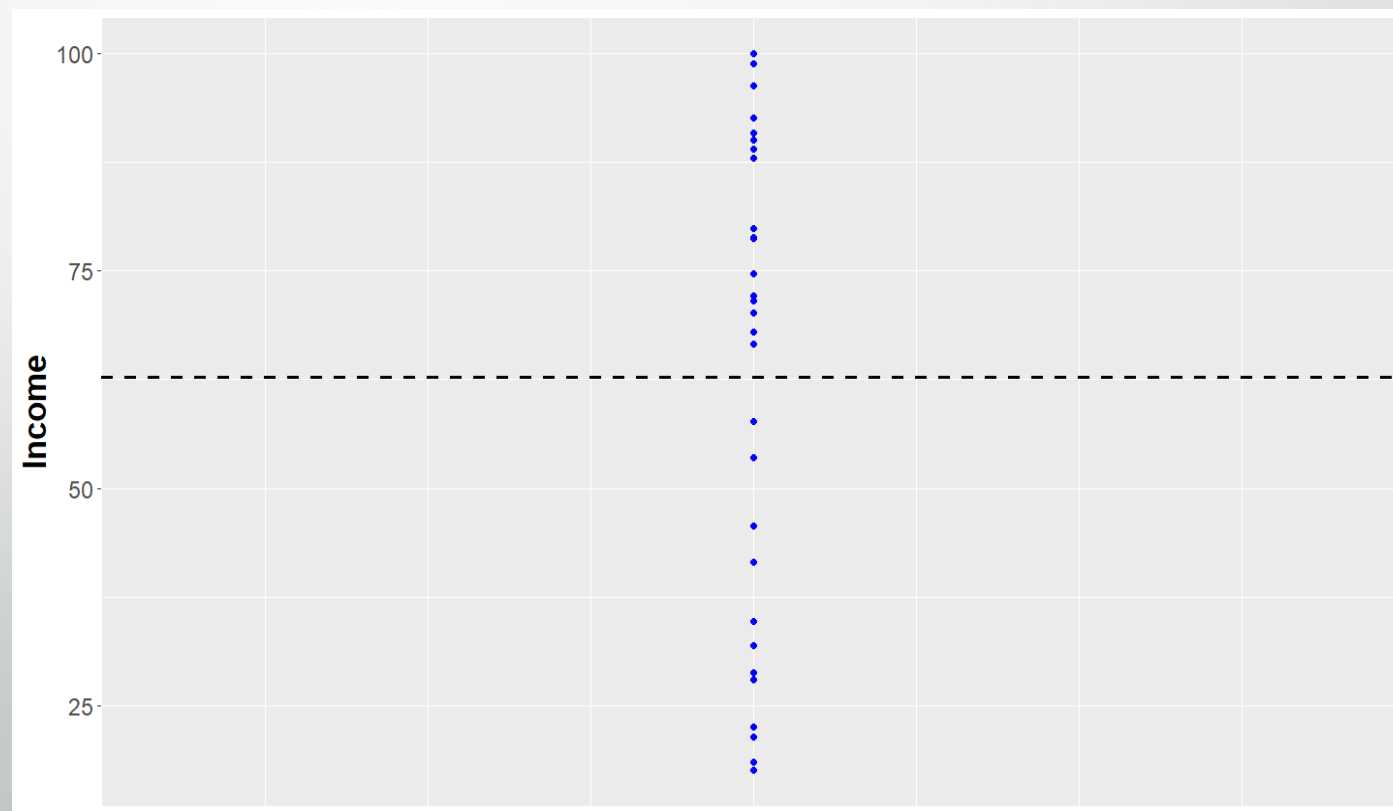


Linear regression with no predictors

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Can we do better?

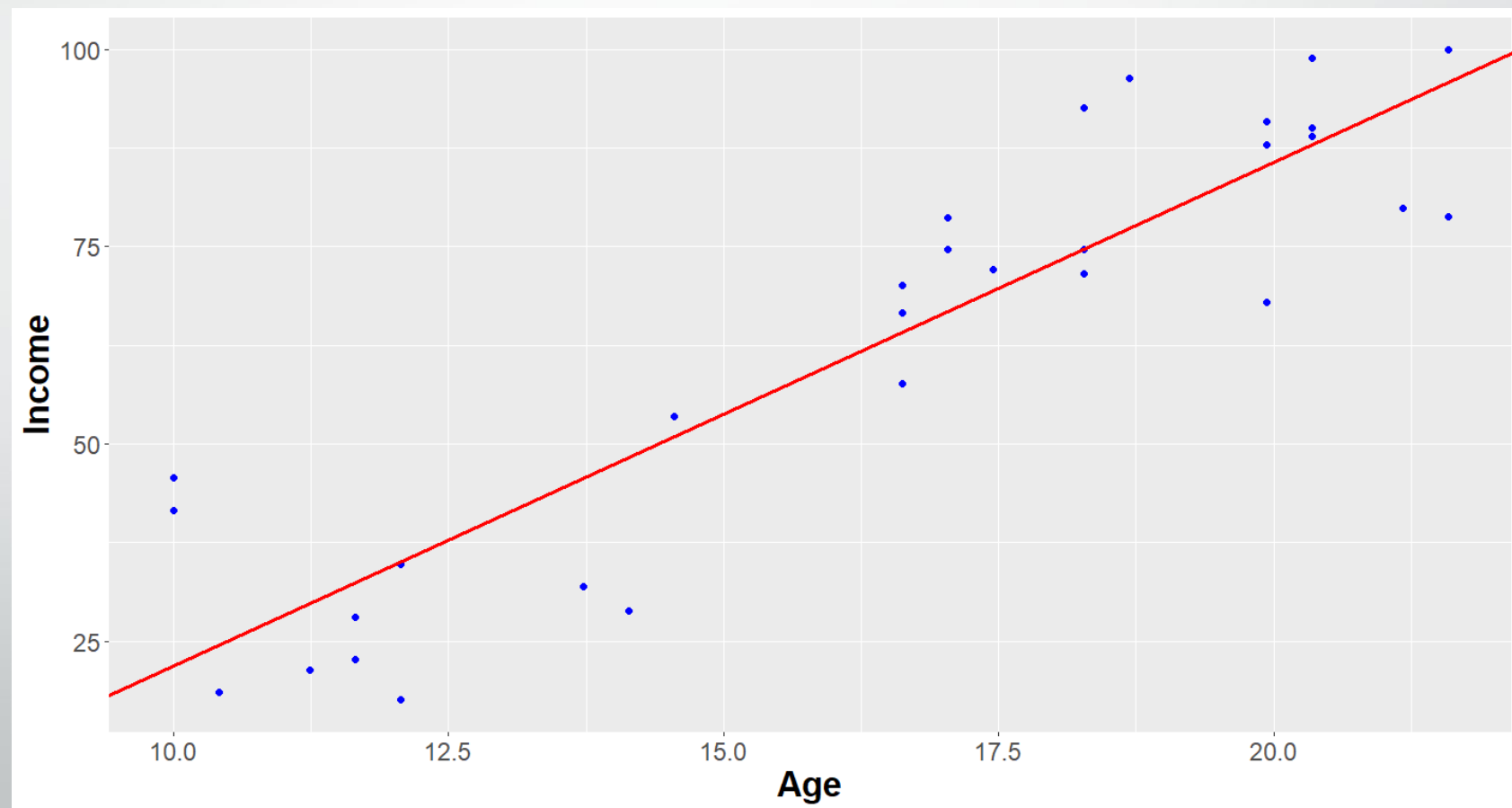


Simple Linear Regression

- Linear regression with one predictor is often referred to as “simple linear regression”.

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Let's look at income as a function of age.



Linear regression notation

- Modeling the outcome variable, Y
 - With no predictors: $Y_i = \bar{Y} + \epsilon_i$
 - With 1 predictor: $Y_i = \beta_0 + \beta_1 X_{1i} + \epsilon_i$
 - With 2 predictors: $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$
 - With 3 predictors: $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \epsilon_i$
 - Etc.
- As long as $f(X)$ is a linear combination of X then it's a Linear Model
 - Linear models can be used to study non-linear relationships so long as predictors form a linear combination

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 - Linear models can be used to study non-linear relationships so long as predictors form a linear combination

What does this look like in R?

Linear Models - 4

- Modeling the outcome variable, Y
 - With no predictors: $Y_i = \bar{Y} + \epsilon_i$ `model_obj <- lm(y ~ 1, data)`
 - With 1 predictor: $Y_i = \beta_0 + \beta_1 X_{1i} + \epsilon_i$ `model_obj <- lm(y ~ var1, data)`
 - With 2 predictors: $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$
 - With 3 predictors: $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \epsilon_i$
- `model_obj <- lm(y ~ var1 + var2, data)`
- `model_obj <- lm(y ~ var1 + var2 + var3, data)`

From data to model

How does our dataset fit into a linear regression model?

Y (Income)	X_1 (Education)	X_2 (Seniority)
99.917	21.586	113.103
92.579	18.276	119.310
34.679	12.069	100.690
$\vdots$	$\vdots$	$\vdots$
74.611	17.034	106.897

$$\begin{pmatrix} 99.917 \\ 92.579 \\ 34.679 \\ \vdots \\ 74.611 \end{pmatrix} = \beta_0 + \beta_1 \times \begin{pmatrix} 21.586 \\ 18.276 \\ 12.069 \\ \vdots \\ 17.034 \end{pmatrix} + \beta_2 \times \begin{pmatrix} 113.103 \\ 119.310 \\ 100.690 \\ \vdots \\ 106.897 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \vdots \\ \epsilon_N \end{pmatrix}$$

More notation!

$$\begin{pmatrix} 99.917 \\ 92.579 \\ 34.679 \\ \vdots \\ 74.611 \end{pmatrix} = \beta_0 + \beta_1 \times \begin{pmatrix} 21.586 \\ 18.276 \\ 12.069 \\ \vdots \\ 17.034 \end{pmatrix} + \beta_2 \times \begin{pmatrix} 113.103 \\ 119.310 \\ 100.690 \\ \vdots \\ 106.897 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \vdots \\ \epsilon_N \end{pmatrix}$$

$$\begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \\ \vdots \\ Y_N \end{pmatrix} = \beta_0 + \beta_1 \times \begin{pmatrix} X_{11} \\ X_{12} \\ X_{13} \\ \vdots \\ X_{1N} \end{pmatrix} + \beta_2 \times \begin{pmatrix} X_{21} \\ X_{22} \\ X_{23} \\ \vdots \\ X_{2N} \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \vdots \\ \epsilon_N \end{pmatrix}$$

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

$$\mathbf{Y} = \beta_0 + \beta_1 \mathbf{X}_1 + \beta_2 \mathbf{X}_2 + \boldsymbol{\epsilon}$$

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

More notation!

$$\begin{pmatrix} 99.917 \\ 92.579 \\ 34.679 \\ \vdots \\ 74.611 \end{pmatrix} = \beta_0 + \beta_1 \times \begin{pmatrix} 21.586 \\ 18.276 \\ 12.069 \end{pmatrix} + \beta_2 \times \begin{pmatrix} 113.103 \\ 119.310 \\ 100.690 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix}$$

$$\mathbf{X} = \begin{pmatrix} X_{11} & X_{12} & \cdots & X_{1P} \\ X_{21} & X_{22} & \cdots & X_{2P} \\ \vdots & \vdots & \ddots & \vdots \\ X_{N1} & X_{N2} & \cdots & X_{NP} \end{pmatrix} = (\mathbf{X}_1 \ \mathbf{X}_2 \ \cdots \ \mathbf{X}_P) = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_N \end{pmatrix}$$

$$\begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \\ \vdots \\ Y_N \end{pmatrix} = \beta_0 + \beta_1 \times \begin{pmatrix} X_{13} \\ \vdots \\ X_{14} \end{pmatrix} + \beta_2 \times \begin{pmatrix} X_{23} \\ \vdots \\ X_{24} \end{pmatrix} + \begin{pmatrix} \epsilon_3 \\ \vdots \\ \epsilon_N \end{pmatrix}$$

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

$$\mathbf{Y} = \beta_0 + \beta_1 \mathbf{X}_1 + \beta_2 \mathbf{X}_2 + \boldsymbol{\epsilon}$$

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Notation review

Given the model: $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$

- Can also be written in matrix form as: $Y = X\beta + \epsilon$

What do we know?

What do we **not** know?

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What do we know?

- We know our data (Y_i and X_i)

What do we **not** know?

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What do we know?

- We know our data (Y_i and X_i)

What do we **not** know?

- $\beta_0, \beta_1, \beta_2$ -- These are our parameters
- We also don't know ϵ , but that's not considered a parameter

Decomposing the linear model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

$$Y = X\beta + \epsilon$$

- What does β_0 mean?

Decomposing the linear model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

$$Y = X\beta + \epsilon$$

- What does β_0 mean?
 - β_0 is the intercept, the expected value of the outcome, Y , when $X = 0$.
- Do we need β_0 ?

Decomposing the linear model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

$$Y = X\beta + \epsilon$$

- What does β_0 mean?
- β_0 is the intercept, the expected value of the outcome, Y , when $X = 0$.
- Do we need β_0 ?
 - Only in rare circumstances can you reasonably remove β_0 from the model.

Decomposing the linear model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

$$Y = X\beta + \epsilon$$

- What about β_1 and β_2 ? What do they represent?

Decomposing the linear model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

$$Y = X\beta + \epsilon$$

- What about β_1 and β_2 ? What do they represent?
 - β_1 and β_2 represent the slope or the expected mean change in the outcome, Y , for every one unit increase of its respective X while holding the other constant
- Do we need β_1 and β_2 ?

Decomposing the linear model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

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- What about β_1 and β_2 ? What do they represent?
 - β_1 and β_2 represent the slope or the expected mean change in the outcome, Y , for every one unit increase of its respective X while holding the other constant
- Do we need β_1 and β_2 ?
 - This depends entirely on the research question(s) and purpose.
 - We'll talk more about this in Class 5!

Decomposing the linear model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

$$Y = X\beta + \epsilon$$

- What does ϵ represent?
 - ϵ represents the error term also known as the residual ($y_i - \hat{y}_i$)
- Do we need it?

Decomposing the linear model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

$$Y = X\beta + \epsilon$$

- What does ϵ represent?
 - ϵ represents the error term also known as the residual ($y_i - \hat{y}_i$)
- Do we need it?
 - Yes! ϵ plays a vital role in regression models as we will see throughout the semester.



Interpreting regression coefficients

Interpreting linear regression coefficients

- Let's fit a model using the income2 data where we model income as a function of education
- We can fit this in R as:

```
model1 <- lm(Income ~ Education, income2)
```

Interpreting linear regression coefficients

```
1 summary(lm(Income ~ Education, income2))
```

Call:

```
lm(formula = Income ~ Education, data = income2)
```

Residuals:

Min	1Q	Median	3Q	Max
-19.568	-8.012	1.474	5.754	23.701

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-41.9166	9.7689	-4.291	0.000192	***
Education	6.3872	0.5812	10.990	1.15e-11	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.93 on 28 degrees of freedom

Multiple R-squared: 0.8118, Adjusted R-squared: 0.8051

F-statistic: 120.8 on 1 and 28 DF, p-value: 1.151e-11

Interpreting linear regression coefficients

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The coefficient for Education is 6.4.

How would you interpret this?

Interpreting linear regression coefficients

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Coefficients:
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The coefficient for Education is 6.4.

How would you interpret this?

- For every one-unit increase in education we expect the mean income to increase by 6.4.

Interpreting linear regression coefficients

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The coefficient for the Intercept is -41.9.

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The coefficient for the Intercept is -41.9.

How would you interpret this?

- The expected income for those where Education = 0 is -41.9.

Does this make sense?



How can we make the intercept more interpretable?

Interpretable intercept via centering

- Centering a numeric variable simply means that we subtract the mean of that variable from each observed value.
- We can create a new centered education variable and refit the model like below.

```
1 income2$edu_cen <- income2$Education - mean(income2$Education)
2 summary(lm(Income ~ edu_cen, income2))
```

Call:

```
lm(formula = Income ~ edu_cen, data = income2)
```

Residuals:

Min	1Q	Median	3Q	Max
-19.568	-8.012	1.474	5.754	23.701

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	62.7447	2.1774	28.82	< 2e-16 ***
edu_cen	6.3872	0.5812	10.99	1.15e-11 ***

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Interpretable intercept via centering

- How would we interpret the intercept in each model?
- What about the slope?

Uncentered

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Interpretable intercept via centering

- How would we interpret the intercept in each model?
 - Hint: The intercept represents Y when the predictor = 0. When does $\text{edu_cen} = 0$?
- What about the slope?

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- What about the slope?

The expected income for those where Education = 0 is -41.9.

Uncentered

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```

The expected income for those with mean level of Education is 62.7

Centered

```
lm(formula = Income ~ edu_cen, data = income2)

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F-statistic: 120.8 on 1 and 28 DF,  p-value: 1.151e-11
```

Interpretable intercept via centering

- How would we interpret the intercept in each model?
 - Hint: The intercept represents Y when the predictor = 0. When does edu_cen = 0?
- What about the slope?

For every one-unit increase in education we expect the mean income to increase by 6.4.

Uncentered

```
Call:
lm(formula = Income ~ Education, data = income)

Residuals:
    Min       1Q   Median       3Q      Max
-19.568  -8.012   1.474   5.754  23.701

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -41.9166    9.7689  -4.291 0.000192 ***
Education       6.3872    0.5812  10.990 1.15e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.93 on 28 degrees of freedom
Multiple R-squared:  0.8118,    Adjusted R-squared:  0.8051
F-statistic: 120.8 on 1 and 28 DF,  p-value: 1.151e-11
```

Centered

```
Call:
lm(formula = Income ~ edu_cen, data = income2)

Residuals:
    Min       1Q   Median       3Q      Max
-19.568  -8.012   1.474   5.754  23.701

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   62.7447    2.1774  28.82 < 2e-16 ***
edu_cen        6.3872    0.5812  10.99 1.15e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.93 on 28 degrees of freedom
Multiple R-squared:  0.8118,    Adjusted R-squared:  0.8051
F-statistic: 120.8 on 1 and 28 DF,  p-value: 1.151e-11
```


Interpretable intercept via centering

Let's look at another example, this time graphically, to see what centering does.

- We'll fit a model exploring weekly wages as a function of age using the [Wage](#) dataset from the ISLR2 package.
 - $E(wage) = \beta_0 + \beta_1(age) + \epsilon$
- How would you interpret the coefficients in the output below?

```
Call:
lm(formula = wage ~ age, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-100.265  -25.115   -6.063   16.601   205.748

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  81.70474    2.84624   28.71  <2e-16 ***
age           0.70728    0.06475   10.92  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.93 on 2998 degrees of freedom
Multiple R-squared:  0.03827,    Adjusted R-squared:  0.03795
F-statistic: 119.3 on 1 and 2998 DF,  p-value: < 2.2e-16
```

Interpretable intercept via centering

Let's look at another example, this time graphically, to see what centering does.

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```

Mean weekly wage for those where age = 0 is 81.7.

Every one-unit increase in age is associated with an increase in the mean expected wage of 0.71.

Interpretable intercept via centering

Let's look at another example, this time graphically, to see what centering does.

- We'll fit a model exploring weekly wages as a function of age using the `Wage` dataset from the `ISLR2` package.
 - $E(wage) = \beta_0 + \beta_1(age) + \epsilon$
- How would you interpret the coefficients in the output below?

```
Call:
lm(formula = wage ~ age, data = Wage)

Residuals:
    Min       1Q   -100.265  -25.115
    1Q      100      25.115   100.265
    Max      100      25.115   100.265

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  81.70474    2.84624   28.71  <2e-16 ***
age           0.70728    0.06475   10.92  <2e-16 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

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Multiple R-squared:  0.03827,    Adjusted R-squared:  0.03795
F-statistic: 119.3 on 1 and 2998 DF,  p-value: < 2.2e-16
```

Do these
interpretations make
sense?

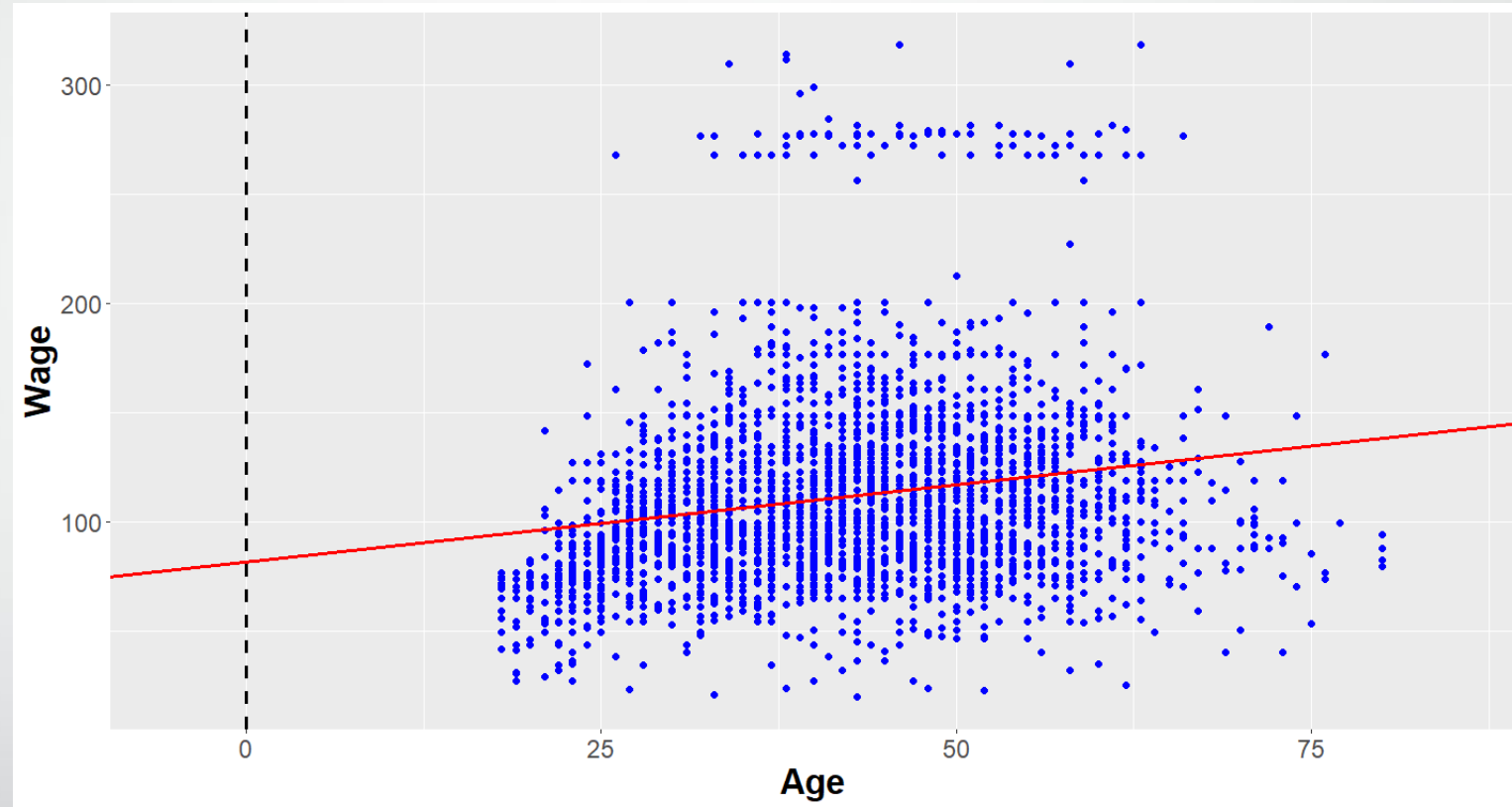
Mean weekly wage for
those where age = 0 is
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in age is associated with
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expected wage of 0.71.

Interpretable intercept via centering

Let's look at this model graphically.

- The red line is the regression line, the blue points are the observed data, and the dashed black vertical line represents the model intercept.

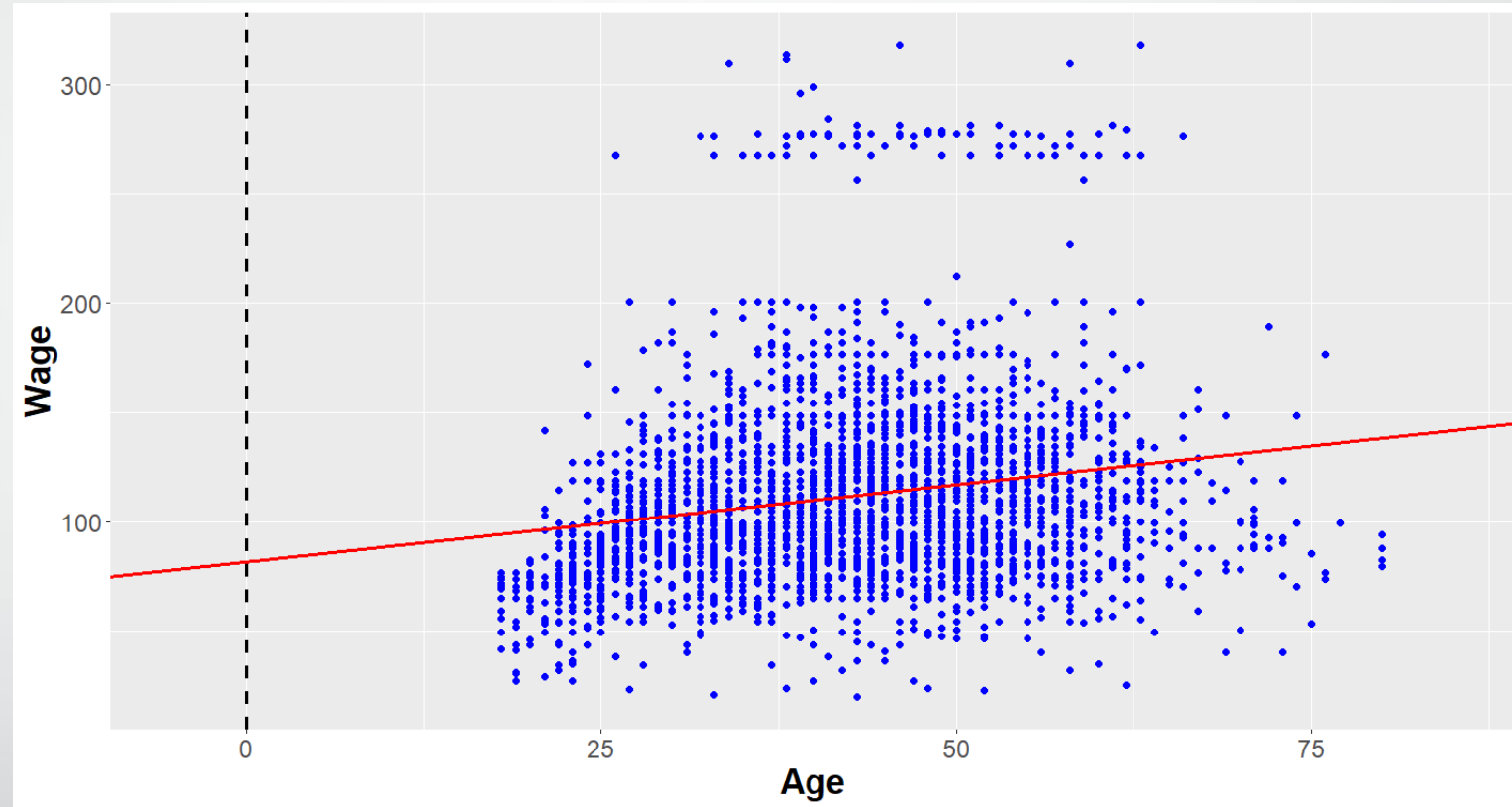


Interpretable intercept via centering

Let's look at this model graphically.

- The red line is the regression line, the blue points are the observed data, and the dashed black vertical line represents the model intercept.

See any potential problems?



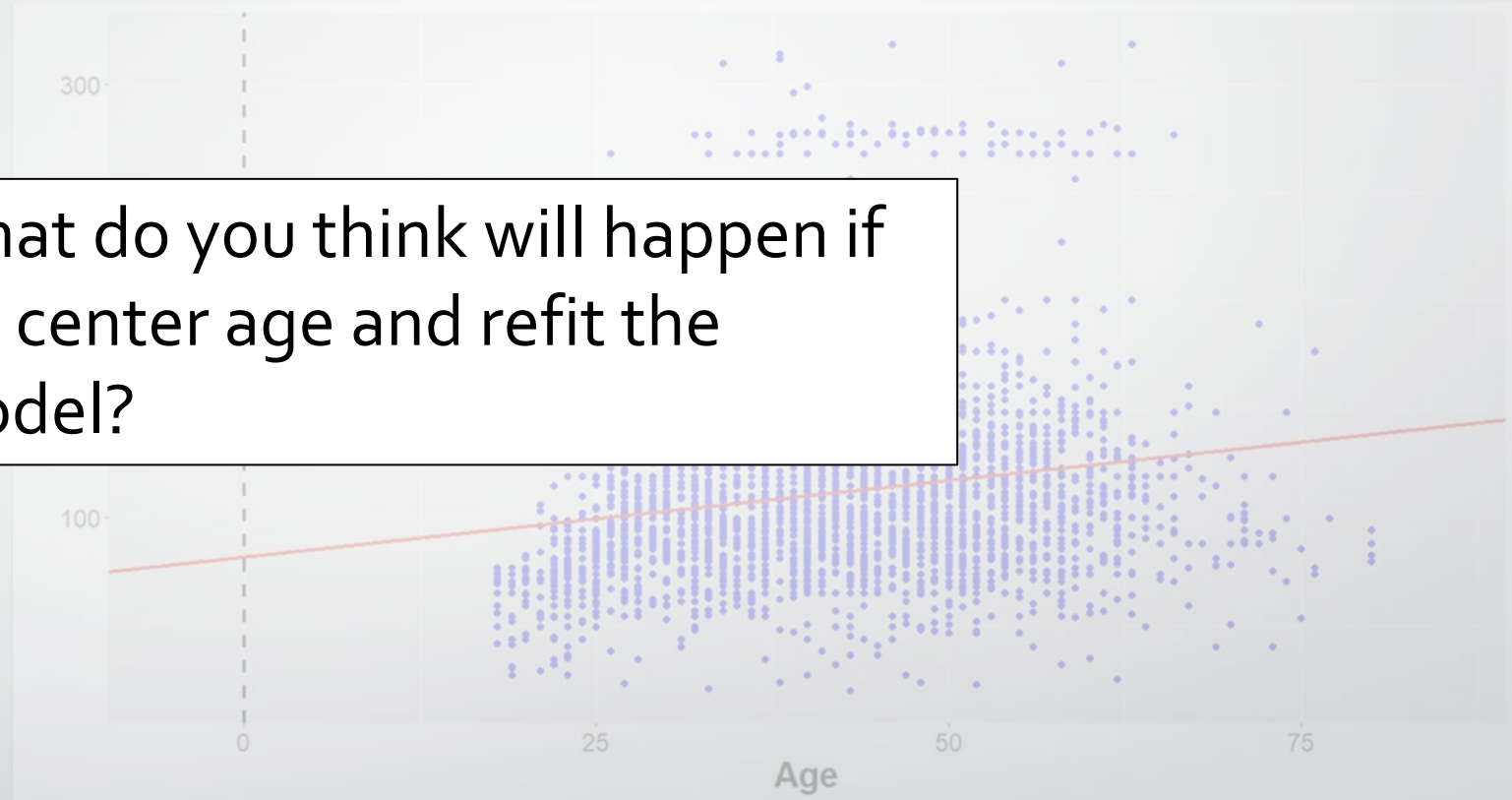
Interpretable intercept via centering

Let's look at this model graphically.

- The red line is the regression line, the blue points are the observed data, and the dashed black vertical line represents the model intercept.

See any potential problems?

What do you think will happen if we center age and refit the model?

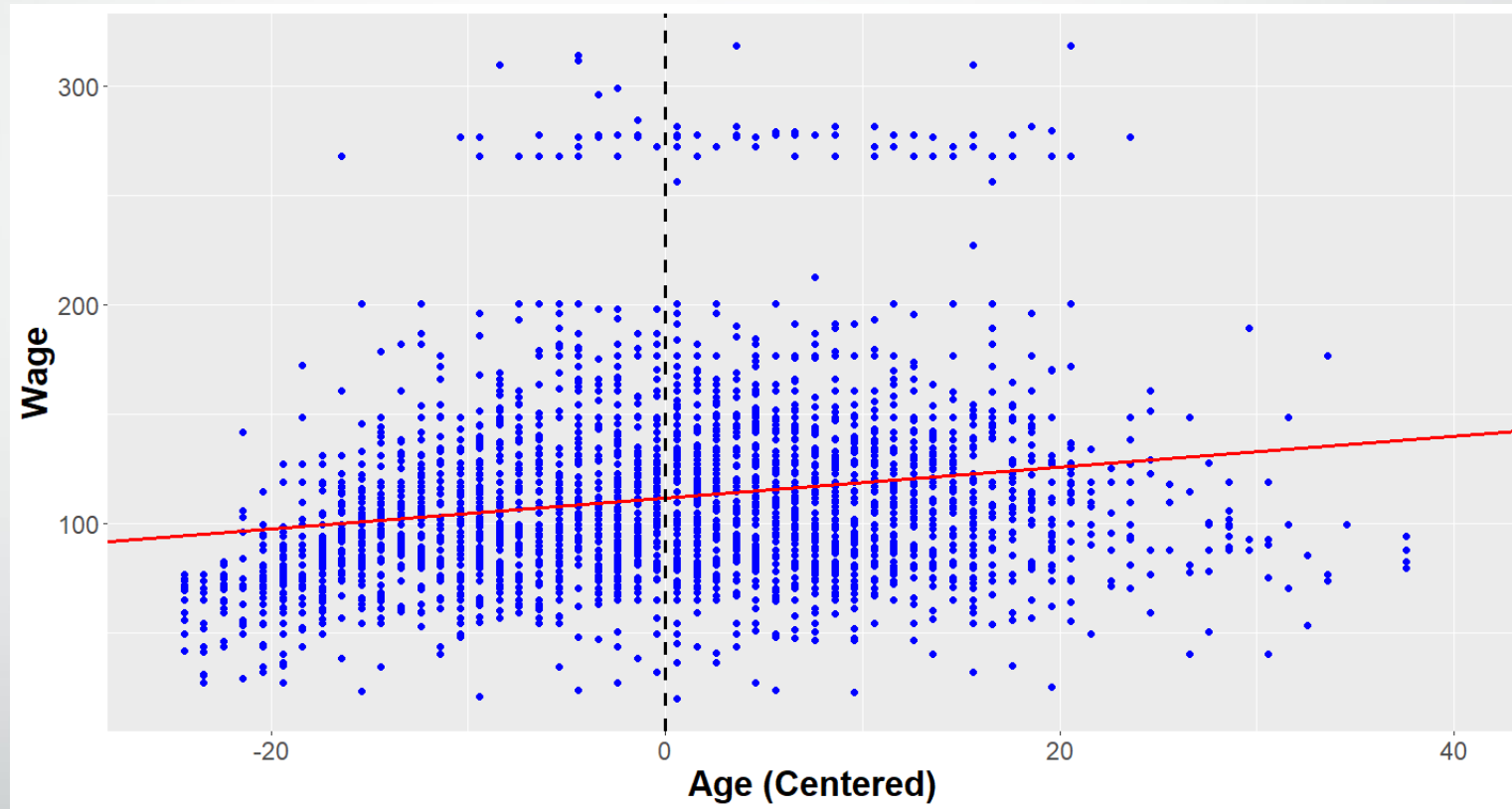


Interpretable intercept via centering

Let's look at a model where age is centered.

What does the intercept represent here?

Is the slope of the regression line different than the uncentered model?



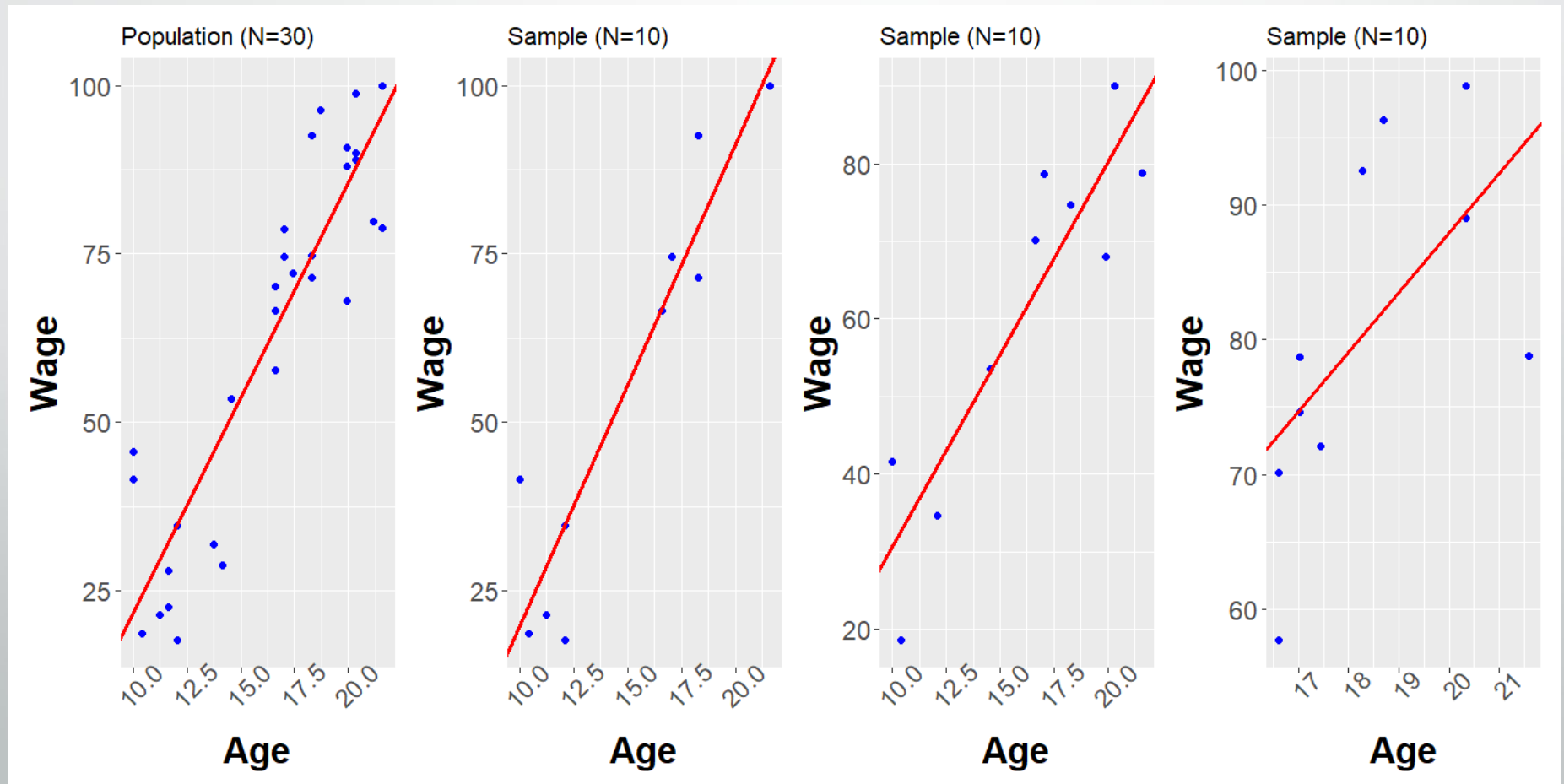
Interpretable intercept via centering

- Can help make intercept more interpretable.
- Doesn't change interpretation of the slope!
 - You're just shifting along the x-axis.

Population vs. Sample

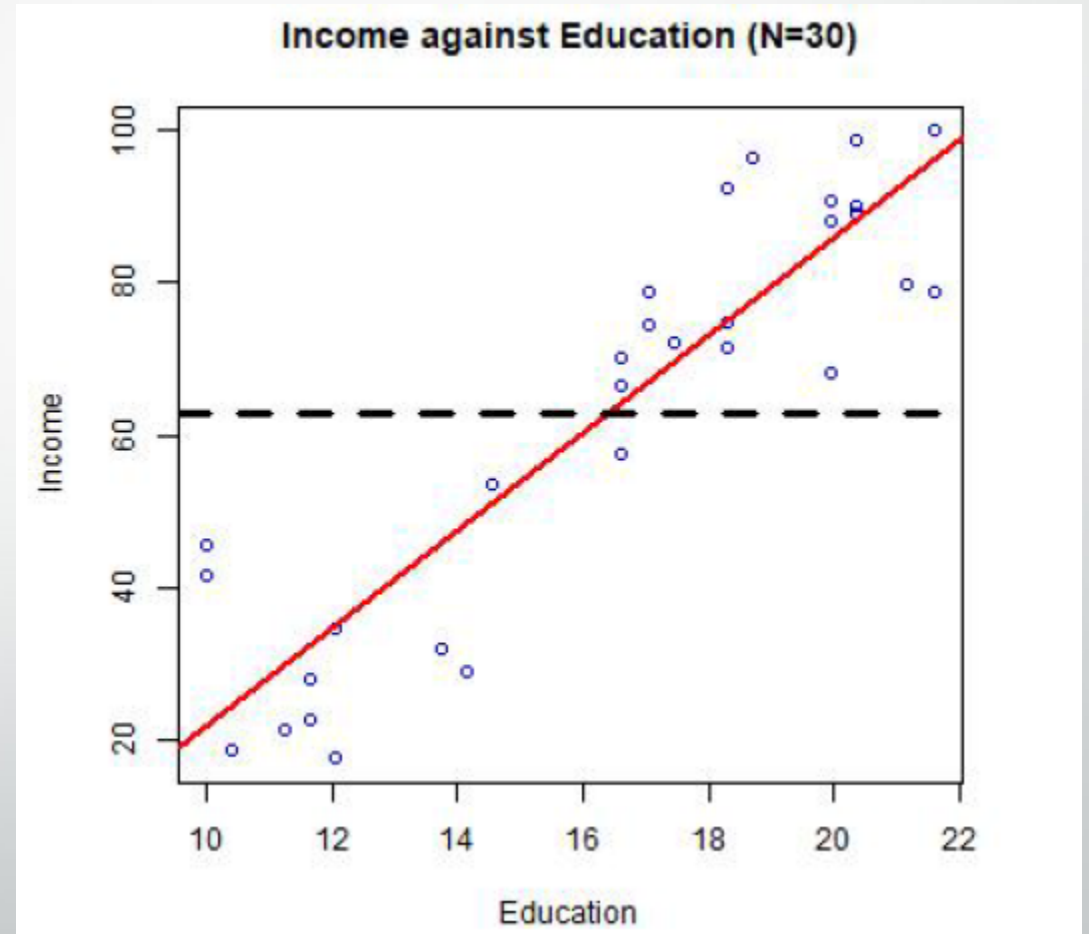
Today we've been treating our data as if they represent the entire population rather than just a sample. This has implications for both the estimates themselves and their notation.

- $\beta_0, \beta_1, \dots, \beta_n$ and $V(\beta_0), V(\beta_1), \dots, V(\beta_n)$
- $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_n$ and $V(\hat{\beta}_0), V(\hat{\beta}_1), \dots, V(\hat{\beta}_n)$



Goal of linear regression model estimation

- Which fit line (red or black) regression lines, do you think does a better job of modeling the relationship between `income` and in these data? `education`



Goal of linear regression model estimation

- The choice in the income and education example was pretty clear, but often it isn't.
- Need a standardized way of finding the regression line that best fits the data
- Our goal is to find unique solutions to $\beta_0, \beta_1, \dots, \beta_n$ that will make $\beta_0 + \beta_1 + \dots + \beta_n$ (i.e., $X\beta$) as close to Y as possible.
 - Put another way – we want to find the smallest ϵ .

We'll discuss how we do this next class!

Goal of Linear Regression Model Estimation-2

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

$$Y = X\beta + \epsilon$$

- Finding unique solutions to $\beta_0, \beta_1, \beta_2$ (i.e., β) that will make $\beta_0 + \beta_1 X_1 + \beta_2 X_2$ (i.e., $X\beta$) as close to Y as possible.
 - Put another way – want to find smallest ϵ with respect to its mean and variance

Next class!

A decorative graphic on the left side of the slide, consisting of several overlapping, angled lines in shades of blue and grey, resembling the collar and lapels of a trench coat.

Common statistical tests are linear
models in trench coats

Statistical tests as linear models

- Showing equivalent linear models
 - One sample t-test
 - Wilcoxon signed-rank

Many more examples in nice post below:

- Pearson correlation, two-sample tests, and many we didn't cover
- <https://lindeloev.github.io/tests-as-linear/>

Statistical tests as linear models

- One sample t-test

```
# One sample t test  
set.seed(894894)  
y <- rnorm(50, 5, 1)  
t.test(y)  
summary(lm(y ~ 1))
```

One Sample t-test

```
data: y  
t = 37.294, df = 49, p-value < 2.2e-16  
alternative hypothesis: true mean is not equal to 0  
95 percent confidence interval:  
 4.803264 5.350396  
sample estimates:  
mean of x  
 5.07683
```

```
Call:  
lm(formula = y ~ 1)
```

```
Residuals:  
      Min       1Q   Median       3Q      Max  
-3.12402 -0.50299  0.06809  0.50362  2.02659
```

```
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)  
(Intercept)   5.0768      0.1361   37.29  <2e-16 ***
```

```
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.9626 on 49 degrees of freedom
```

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Statistical tests as linear models

- Wilcoxon signed-rank

```
# wilcoxon signed rank test
signed_rank = function(x) sign(x) * rank(abs(x))

set.seed(894894)
y = c(rnorm(15), exp(rnorm(15)), runif(20, min = -3, max = 0))
wilcox.test(y)
summary(lm(signed_rank(y) ~ 1))
t.test(signed_rank(y))
```

wilcoxon signed rank test with continuity correction

```
data: y
V = 1091, p-value = 0.3769
alternative hypothesis: true location is not equal to 0
```

One Sample t-test

```
data: signed_rank(y)
t = -0.88523, df = 69, p-value = 0.3791
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
-14.083348  5.426205
sample estimates:
mean of x
-4.328571
```

```
Call:
lm(formula = signed_rank(y) ~ 1)
```

Residuals:

Min	1Q	Median	3Q	Max
-61.671	-32.421	-5.171	33.829	74.329

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4.329	4.890	-0.885	0.379

Residual standard error: 40.91 on 69 degrees of freedom

Statistical tests as linear models

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